

Mathematical Induction

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Statements indexed by integers

In this lecture, let

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

denote the positive integers.

Many theorems assert that a property $P(n)$ holds for every $n \in \mathbb{N}$.

Examples include formulas for sums, inequalities, divisibility statements, and claims about algorithms after n iterations.

Checking finitely many values cannot prove a statement for every natural number.

Finite sums

The notation

$$\sum_{i=1}^n a_i$$

means

$$a_1 + a_2 + \cdots + a_n.$$

The variable i is the **index of summation**; 1 and n are the lower and upper limits.

For example,

$$\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2.$$

Changing the range

Another example is

$$\sum_{i=7}^9 (i^2 + i) = (7^2 + 7) + (8^2 + 8) + (9^2 + 9).$$

If $A = \{1, 2, 4, 15\}$, then

$$\sum_{i \in A} i^2 = 1^2 + 2^2 + 4^2 + 15^2.$$

A sum over an empty index set is an **empty sum**. By convention, its value is 0.

A formula for a sum

Consider the proposition

$$P(n) : \sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

It is easy to check $P(1)$, $P(2)$, and $P(3)$.

These checks provide evidence, but they do not prove

$$(\forall n \in \mathbb{N})P(n).$$

The useful observation is that the sum for $n+1$ contains the sum for n .

From one integer to the next

Suppose that, for some integer $k \geq 1$,

$$\sum_{i=1}^k i = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \left(\sum_{i=1}^k i \right) + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}.\end{aligned}$$

Thus $P(k)$ implies $P(k+1)$.

The induction chain

Recall: modus ponens. If A and $A \Rightarrow B$ are both true, then B is true:

$$A, \quad A \Rightarrow B \quad \therefore B.$$

Suppose we know

$$P(1) \quad \text{and} \quad (\forall k \geq 1)(P(k) \Rightarrow P(k+1)).$$

Then repeated modus ponens gives

$$P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow P(4) \Rightarrow \dots$$

The principle of mathematical induction formalizes the conclusion that every $P(n)$ is true.

Principle of mathematical induction

Let $P(n)$ be a property defined for every integer $n \geq n_0$.

To prove $(\forall n \geq n_0)P(n)$, it is enough to prove:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every $k \geq n_0$,

$$P(k) \Rightarrow P(k + 1).$$

In the inductive step, the assumption $P(k)$ is the **induction hypothesis**.

What the inductive step proves

The inductive step does not assume that $P(k)$ is true for every k .
It proves the universal implication

$$(\forall k \geq n_0)(P(k) \Rightarrow P(k+1)).$$

Inside that proof, k is arbitrary and $P(k)$ is assumed temporarily.
The base case supplies the first true proposition; the inductive step propagates truth to every later integer.

Why do we need a base case?

The inductive step proves that the implication

$$P(k) \Rightarrow P(k + 1)$$

is true. It does not prove that $P(k)$ itself is true.

The base case establishes the first true premise, $P(n_0)$.

We can then apply modus ponens:

$$P(n_0), \quad P(n_0) \Rightarrow P(n_0 + 1) \quad \therefore \quad P(n_0 + 1).$$

Repeating this valid argument produces

$$P(n_0), P(n_0 + 1), P(n_0 + 2), \dots$$

Without the base case, the induction chain has no true starting premise.

The sum of the first integers

Theorem

For every integer $n \geq 1$,

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

Define

$$P(n) : \sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

We prove $P(n)$ for all $n \geq 1$ by induction.

Base case

For $n = 1$,

$$\sum_{i=1}^1 i = 1$$

and

$$\frac{1(1+1)}{2} = 1.$$

Therefore $P(1)$ is true.

A base case must match the smallest integer covered by the theorem.

Inductive step

Let $k \geq 1$ be arbitrary, and assume the induction hypothesis

$$\sum_{i=1}^k i = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \left(\sum_{i=1}^k i \right) + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2} \\ &= \frac{(k+1)((k+1)+1)}{2}.\end{aligned}$$

Thus $P(k+1)$ is true.

Conclusion

We have established:

- ▶ $P(1)$;
- ▶ for every $k \geq 1$, $P(k) \Rightarrow P(k + 1)$.

By the principle of mathematical induction,

$$(\forall n \geq 1)P(n).$$

Therefore,

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

for every integer $n \geq 1$.

Common errors

1. Checking several values and calling the checks a proof.
2. Omitting the base case.
3. Assuming $P(k + 1)$ instead of assuming $P(k)$.
4. Proving the algebraic identity for one particular k rather than an arbitrary k .
5. Using the induction hypothesis without identifying the step where it is used.

Summary

- ▶ Induction proves properties indexed by consecutive integers.
- ▶ The base case starts the chain of implications.
- ▶ The inductive step proves $P(k) \Rightarrow P(k + 1)$ for arbitrary k .
- ▶ The induction hypothesis may be used only within the inductive step.
- ▶ A complete proof states the property, base case, inductive step, and conclusion.